

Scientific Evolution Journal

Project Genesis Research Repository

Generated: July 2026

Scientific Evolution Journal

Project Genesis Automated Evolutionary Analysis & Telemetry Log

This journal catalogs major genetic shifts, physiological dynamics, and cognitive adaptations across the simulation's epochs.

Year 0 (Tick 1)

- \textbf{Demographics}: 16 agents alive / 16 total. Maximum Generation: Gen 0.
- \textbf{Climate Phase}: \texttt{Temperate} (Temp Offset: +0.0C, Water Mult: 1.00x)
- \textbf{Top Lineages}: Founder #0 (1 alive), Founder #1 (1 alive), Founder #2 (1 alive), Founder #3 (1 alive), Founder #4 (1 alive)

Genetic Means Shift

Gene / Trait	Starting Mean	Current Mean	Shift	Variance
g_metabolism	0.647	0.647	+0.000	0.0499
g_thermoregulation	0.479	0.479	+0.000	0.1137
g_vision	0.440	0.440	+0.000	0.0713
g_mobility	0.589	0.589	+0.000	0.0580
g_memory_fidelity	0.497	0.497	+0.000	0.0749
g_planning	0.498	0.498	+0.000	0.0690
g_novelty_seeking	0.467	0.467	+0.000	0.0681
g_social_proximity	0.583	0.583	+0.000	0.0554
g_aggression	0.573	0.573	+0.000	0.0970
g_resource_sharing	0.598	0.598	+0.000	0.0715
g_risk_sensitivity	0.619	0.619	+0.000	0.0609
g_resilience	0.702	0.702	+0.000	0.0428
g_longevity	0.555	0.555	+0.000	0.0730
g_learning_rate	0.464	0.464	+0.000	0.0700

Scientific Insights & Discoveries

Civilization has achieved high prediction accuracy.

- \textbf{Category}: \texttt{Cognitive Evolution}
- \textbf{Evidence}: Living population averaged 100.0% successful spatial resource predictions.
- \textbf{Explanatory Hypothesis}: Neural predictor networks have successfully calibrated against the local environment. Agents are anticipating resources rather than reacting blindly.
- \textbf{Analytical Confidence}: \texttt{90}\% (based on colony congruency)

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Genetic Means Shift

Gene / Trait	Starting Mean	Current Mean	Shift	Variance
g_metabolism	0.576	0.576	+0.000	0.0908
g_thermoregulation	0.490	0.490	+0.000	0.0856
g_vision	0.524	0.524	+0.000	0.0965
g_mobility	0.472	0.472	+0.000	0.0899
g_memory_fidelity	0.531	0.531	+0.000	0.0867
g_planning	0.522	0.522	+0.000	0.1081
g_novelty_seeking	0.575	0.575	+0.000	0.0863
g_social_proximity	0.354	0.354	+0.000	0.1277
g_aggression	0.528	0.528	+0.000	0.0977
g_resource_sharing	0.502	0.502	+0.000	0.0586
g_risk_sensitivity	0.470	0.470	+0.000	0.0785
g_resilience	0.531	0.531	+0.000	0.0801
g_longevity	0.404	0.404	+0.000	0.0593
g_learning_rate	0.501	0.501	+0.000	0.0980

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Genetic Means Shift

Gene / Trait	Starting Mean	Current Mean	Shift	Variance
g_metabolism	0.582	0.582	+0.000	0.0865
g_thermoregulation	0.521	0.521	+0.000	0.1022
g_vision	0.460	0.460	+0.000	0.0715
g_mobility	0.491	0.491	+0.000	0.0624
g_memory_fidelity	0.422	0.422	+0.000	0.0758
g_planning	0.573	0.573	+0.000	0.0839
g_novelty_seeking	0.563	0.563	+0.000	0.0888
g_social_proximity	0.468	0.468	+0.000	0.1190
g_aggression	0.642	0.642	+0.000	0.0615
g_resource_sharing	0.514	0.514	+0.000	0.0795
g_risk_sensitivity	0.588	0.588	+0.000	0.0802
g_resilience	0.585	0.585	+0.000	0.0704
g_longevity	0.596	0.596	+0.000	0.1286
g_learning_rate	0.543	0.543	+0.000	0.0788

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g_vision	0.543	0.543	+0.000	0.0919
g_mobility	0.525	0.525	+0.000	0.0899

Gene / Trait	Starting Mean	Current Mean	Shift	Variance
g_memory_fidelity	0.499	0.499	+0.000	0.1008
g_planning	0.454	0.454	+0.000	0.1015
g_novelty_seeking	0.434	0.434	+0.000	0.0654
g_social_proximity	0.459	0.459	+0.000	0.0597
g_aggression	0.504	0.504	+0.000	0.1033
g_resource_sharing	0.372	0.372	+0.000	0.0751
g_risk_sensitivity	0.578	0.578	+0.000	0.0523
g_resilience	0.558	0.558	+0.000	0.0752
g_longevity	0.437	0.437	+0.000	0.0872
g_learning_rate	0.551	0.551	+0.000	0.0686

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g_metabolism	0.567	0.567	+0.000	0.0881
g_thermoregulation	0.427	0.427	+0.000	0.0664
g_vision	0.426	0.426	+0.000	0.0610
g_mobility	0.470	0.470	+0.000	0.1009
g_memory_fidelity	0.548	0.548	+0.000	0.0674
g_planning	0.453	0.453	+0.000	0.0802
g_novelty_seeking	0.561	0.561	+0.000	0.0828
g_social_proximity	0.467	0.467	+0.000	0.0484
g_aggression	0.404	0.404	+0.000	0.0850
g_resource_sharing	0.583	0.583	+0.000	0.0490
g_risk_sensitivity	0.582	0.582	+0.000	0.0663
g_resilience	0.620	0.620	+0.000	0.0520
g_longevity	0.507	0.507	+0.000	0.0641
g_learning_rate	0.465	0.465	+0.000	0.0942

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g_thermoregulation	0.609	0.609	+0.000	0.0899
g_vision	0.574	0.574	+0.000	0.0841
g_mobility	0.565	0.565	+0.000	0.0721
g_memory_fidelity	0.563	0.563	+0.000	0.1119
g_planning	0.469	0.469	+0.000	0.0770
g_novelty_seeking	0.435	0.435	+0.000	0.0724
g_social_proximity	0.421	0.421	+0.000	0.0890
g_aggression	0.479	0.479	+0.000	0.0914
g_resource_sharing	0.445	0.445	+0.000	0.0778
g_risk_sensitivity	0.364	0.364	+0.000	0.1027
g_resilience	0.373	0.373	+0.000	0.0836
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g_vision	0.501	0.501	+0.000	0.0852
g_mobility	0.638	0.638	+0.000	0.0608
g_memory_fidelity	0.530	0.530	+0.000	0.0763
g_planning	0.524	0.524	+0.000	0.0988
g_novelty_seeking	0.399	0.399	+0.000	0.0630
g_social_proximity	0.491	0.491	+0.000	0.0688
g_aggression	0.479	0.479	+0.000	0.0548
g_resource_sharing	0.512	0.512	+0.000	0.0732
g_risk_sensitivity	0.552	0.552	+0.000	0.0797
g_resilience	0.447	0.447	+0.000	0.0999
g_longevity	0.459	0.459	+0.000	0.0437
g_learning_rate	0.495	0.495	+0.000	0.0952

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Genetic Means Shift

Gene / Trait	Starting Mean	Current Mean	Shift	Variance
g_metabolism	0.420	0.420	+0.000	0.0940
g_thermoregulation	0.586	0.586	+0.000	0.1194
g_vision	0.501	0.501	+0.000	0.0852
g_mobility	0.638	0.638	+0.000	0.0608
g_memory_fidelity	0.530	0.530	+0.000	0.0763
g_planning	0.524	0.524	+0.000	0.0988
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Genetic Means Shift

Gene / Trait	Starting Mean	Current Mean	Shift	Variance
g_metabolism	0.516	0.516	+0.000	0.0883
g_thermoregulation	0.593	0.593	+0.000	0.0567
g_vision	0.567	0.567	+0.000	0.1006
g_mobility	0.289	0.289	+0.000	0.0499
g_memory_fidelity	0.570	0.570	+0.000	0.0884
g_planning	0.542	0.542	+0.000	0.0694
g_novelty_seeking	0.631	0.631	+0.000	0.0474
g_social_proximity	0.514	0.514	+0.000	0.0543
g_aggression	0.507	0.507	+0.000	0.1024
g_resource_sharing	0.511	0.511	+0.000	0.0876
g_risk_sensitivity	0.545	0.545	+0.000	0.0622
g_resilience	0.569	0.569	+0.000	0.0668
g_longevity	0.400	0.400	+0.000	0.0661
g_learning_rate	0.424	0.424	+0.000	0.0676

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Gene / Trait	Starting Mean	Current Mean	Shift	Variance
g_metabolism	0.578	0.578	+0.000	0.0878
g_thermoregulation	0.461	0.461	+0.000	0.0484
g_vision	0.376	0.376	+0.000	0.0611
g_mobility	0.561	0.561	+0.000	0.0989

Gene / Trait	Starting Mean	Current Mean	Shift	Variance
g_memory_fidelity	0.531	0.531	+0.000	0.0987
g_planning	0.494	0.494	+0.000	0.0833
g_novelty_seeking	0.567	0.567	+0.000	0.0564
g_social_proximity	0.576	0.576	+0.000	0.0746
g_aggression	0.475	0.475	+0.000	0.0688
g_resource_sharing	0.545	0.545	+0.000	0.0696
g_risk_sensitivity	0.559	0.559	+0.000	0.0933
g_resilience	0.611	0.611	+0.000	0.0580
g_longevity	0.530	0.530	+0.000	0.0602
g_learning_rate	0.437	0.437	+0.000	0.0763

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g_metabolism	0.563	0.563	+0.000	0.0820
g_thermoregulation	0.466	0.466	+0.000	0.0816
g_vision	0.534	0.534	+0.000	0.0487
g_mobility	0.611	0.611	+0.000	0.1101
g_memory_fidelity	0.474	0.474	+0.000	0.0718
g_planning	0.488	0.488	+0.000	0.1214
g_novelty_seeking	0.460	0.460	+0.000	0.0844
g_social_proximity	0.472	0.472	+0.000	0.0704
g_aggression	0.497	0.497	+0.000	0.0497
g_resource_sharing	0.496	0.496	+0.000	0.0768
g_risk_sensitivity	0.471	0.471	+0.000	0.0689
g_resilience	0.375	0.375	+0.000	0.0590
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g_metabolism	0.563	0.563	+0.000	0.0820
g_thermoregulation	0.466	0.466	+0.000	0.0816
g_vision	0.534	0.534	+0.000	0.0487
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g_memory_fidelity	0.474	0.474	+0.000	0.0718
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